

Sets in Go: past, present and future

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My best work (so far): Fluent Python



- Published in 9 languages, 2 editions (2015, 2022)
- A deep dive into idiomatic Python, exploring the design of the language and its standard library

The Typing Map

The four typing approaches depicted in **Figure 13-1** are complementary: they have different pros and cons. It doesn't make sense to dismiss any of them.

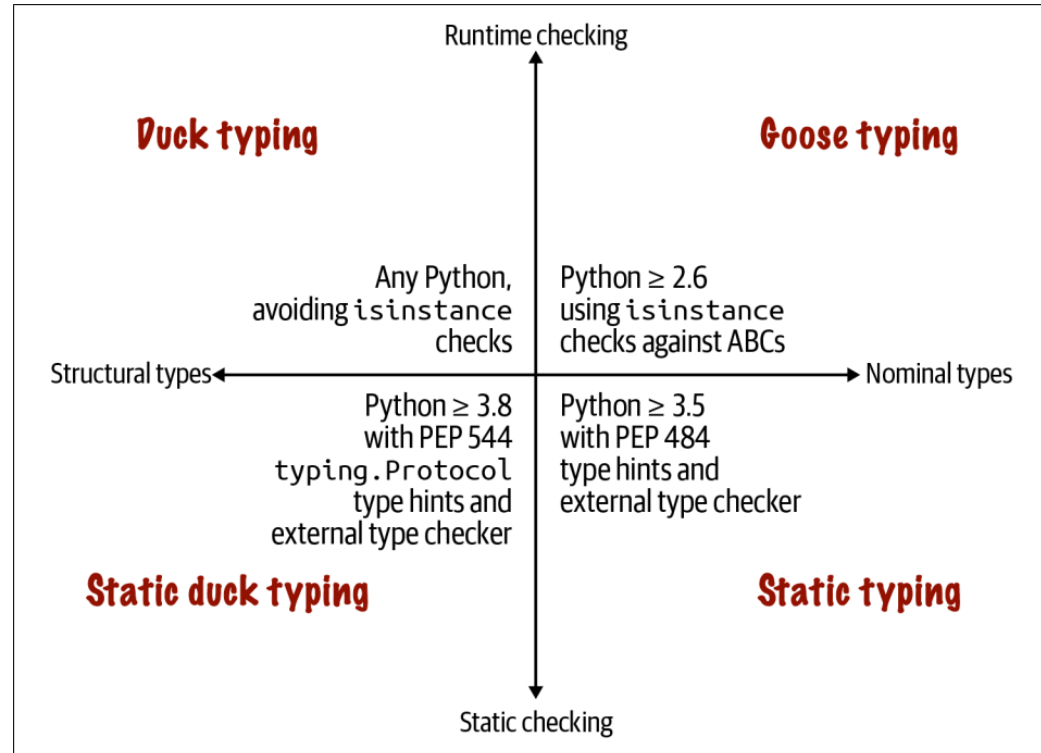


Figure 13-1. The top half describes runtime type checking approaches using just the Python interpreter; the bottom requires an external static type checker such as MyPy or an IDE like PyCharm. The left quadrants cover typing based on the object's structure—i.e., the methods provided by the object, regardless of the name of its class or super-classes; the right quadrants depend on objects having explicitly named types: the name of the object's class, or the name of its superclasses.

Typing Approaches in Popular Languages

Figure 13-8 is a variation of the Typing Map (Figure 13-1) with the names of a few popular languages that support each of the typing approaches.

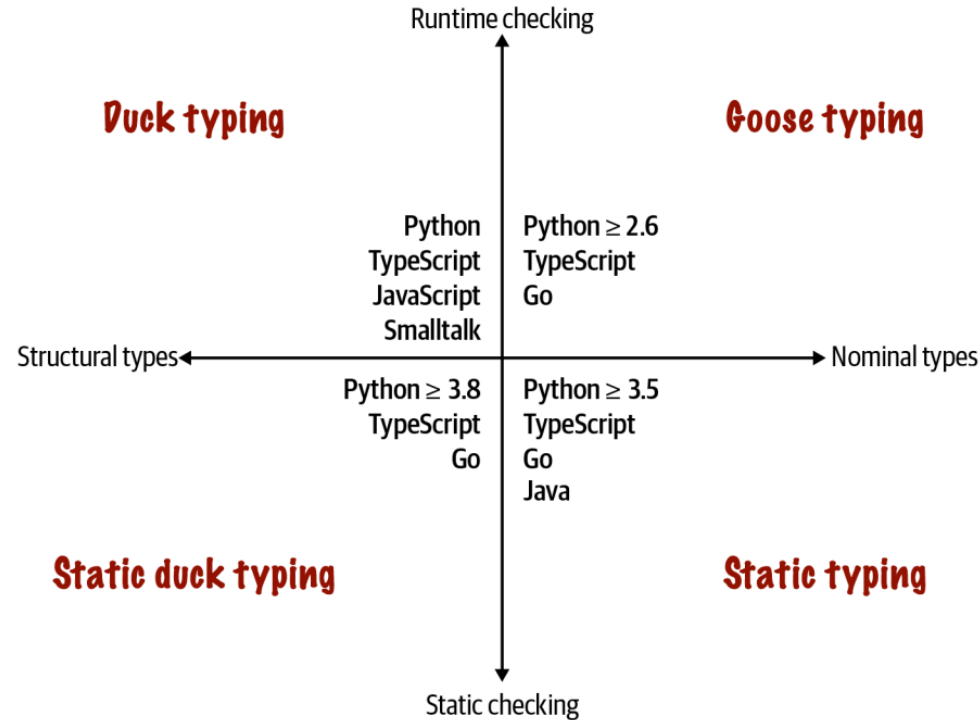


Figure 13-8. Four approaches to type checking and some languages that support them.

TypeScript and Python ≥ 3.8 are the only languages in my small and arbitrary sample that support all four approaches.

Go is clearly a statically typed language in the Pascal tradition, but it pioneered static duck typing—at least among languages that are widely used today. I also put Go in the goose typing quadrant because of its type assertions, which allow checking and adapting to different types at runtime.

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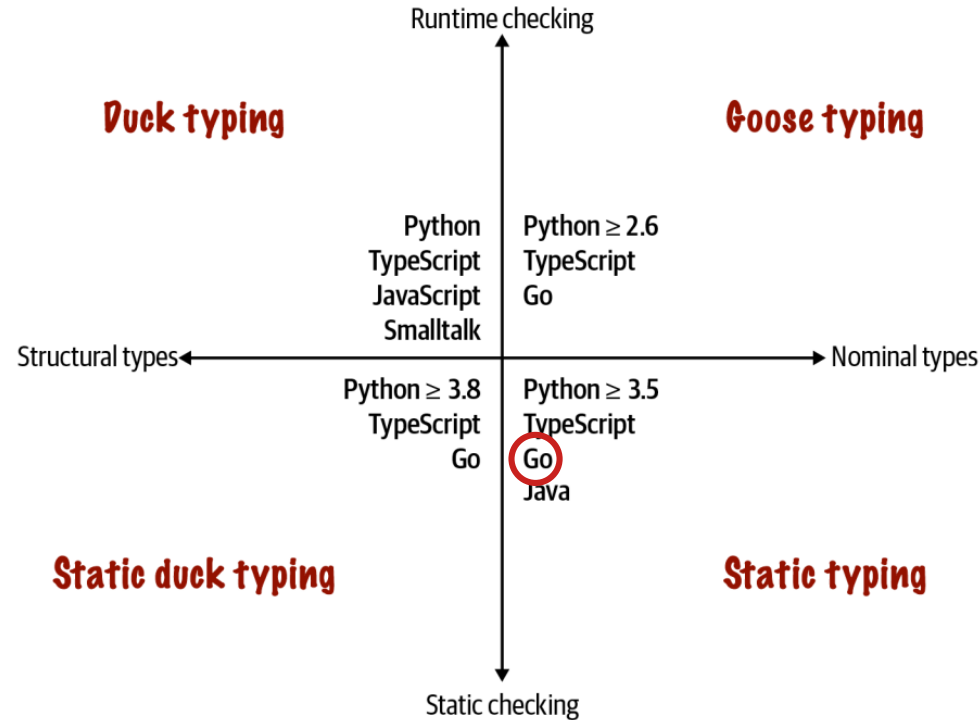


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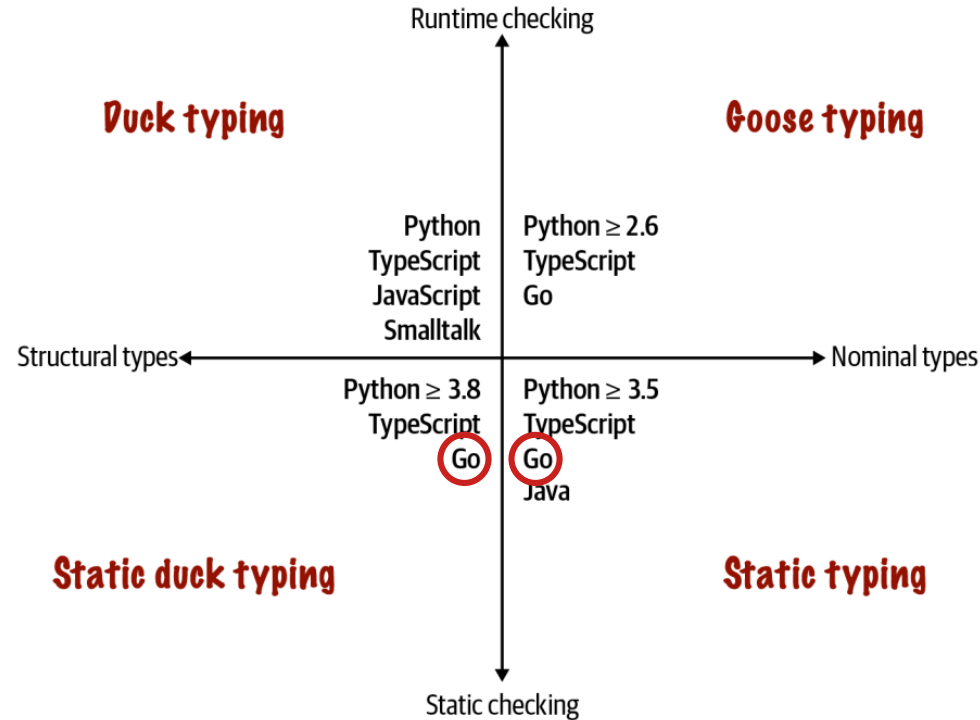


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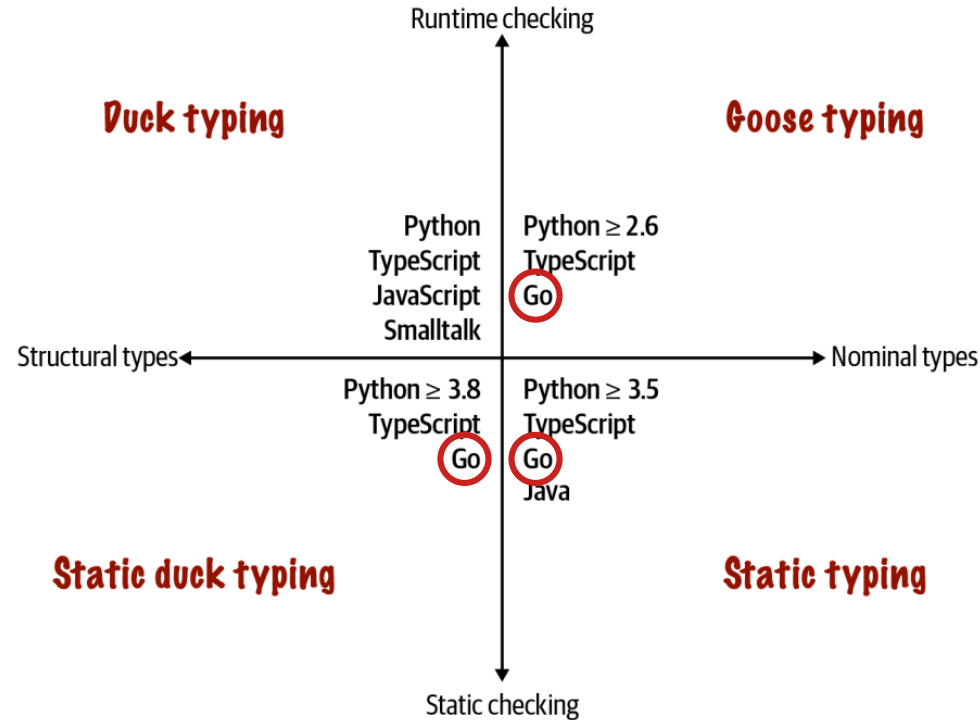


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About me

- Member of the Bartz v. Anthropic class action
- Principal consultant at Thoughtworks (2015-2023)
- Instructional designer at Oficina Turing
- Co-founder of Garoa Hacker Clube,
a hackerspace in São Paulo, Brasil (since 2010)

A tale of two talks

- In 2018 I presented "Set Practice" at GopherCon Brasil:
 - Pitch: "Why and how to implement sets in Go"
- This is a radical update, covering the set types in the collections package for Go 1.28

vintage
2018

ThoughtWorks®

construção e uso

PRÁTICA DE CONJUNTOS

Porquê e como implementar um tipo Set na linguagem Go.



Luciano Ramalho
@ramalhoorg

USE CASE #1

**display product if all
words in the query appear
in the product
description.**

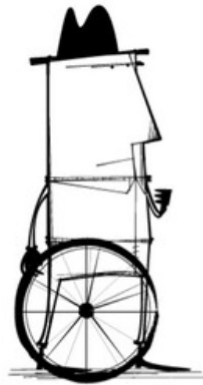
HAND-ROLLED SOLUTION #1

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I've written code like this in Go, which lacks built-in sets:

```
func ContainsAll(slice, subslice []string) bool {  
    for _, needle := range subslice {  
        found := false  
        for _, elem := range slice {  
            if needle == elem {  
                found = true  
                break  
            }  
        }  
        if !found {  
            return false  
        }  
    }  
    return true  
}
```

What if...



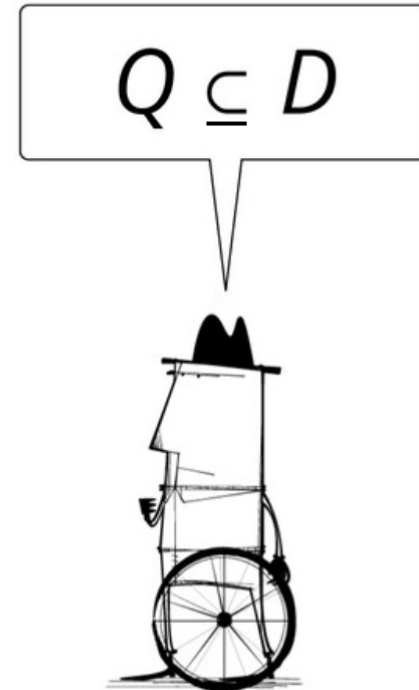
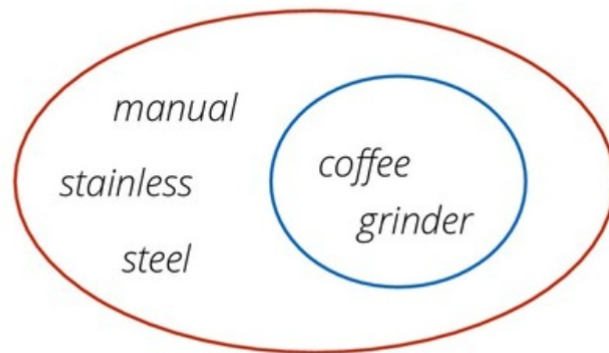
Later! I am too busy
coding nested loops!



USE CASE #1

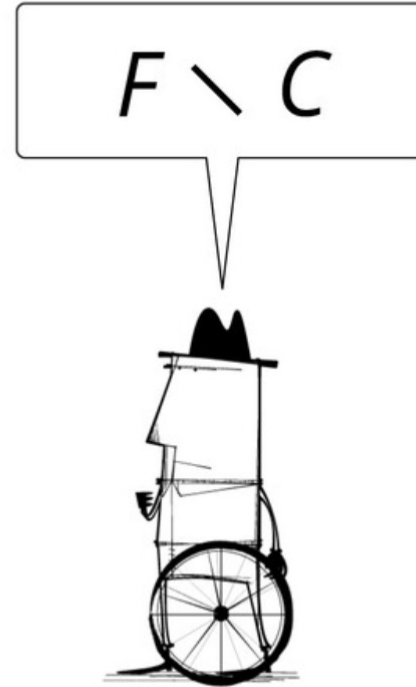
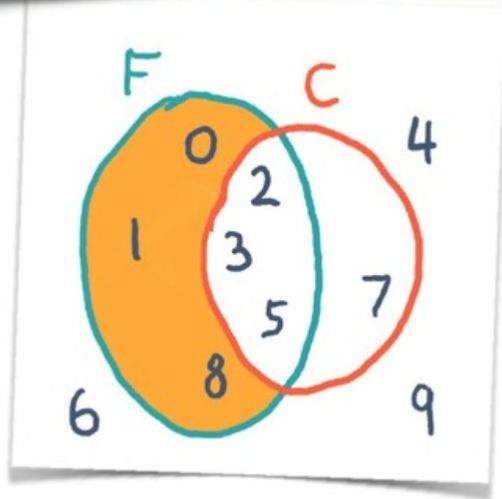
www.workcompass.com/

display product if all
words in the query appear
in the product
description.



USE CASE #2

Mark all products previously favorited, except those already in the shopping cart.



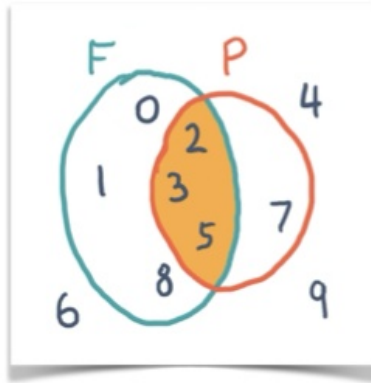


Sets and Logic

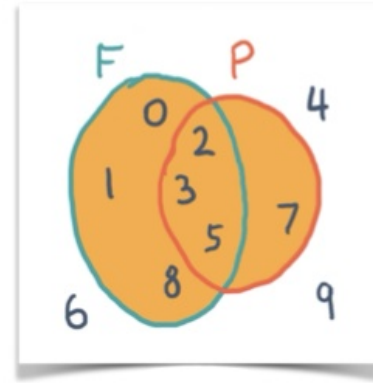
Nobody has yet discovered a branch of mathematics that has successfully resisted formalization into set theory.

Thomas Forster
Logic Induction and Sets
p. 167

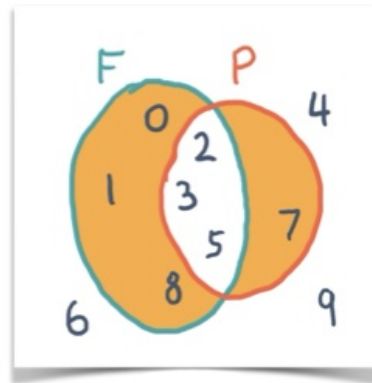
SOME SET OPERATIONS AND LOGIC EQUIVALENTS



Intersection
a.k.a. logical
conjunction



Union
a.k.a. logical
disjunction



**Symmetric
difference**
a.k.a. negation
of conjunction

LOGIC CONJUNCTION IS INTERSECTION

x belongs to the intersection of A with B.

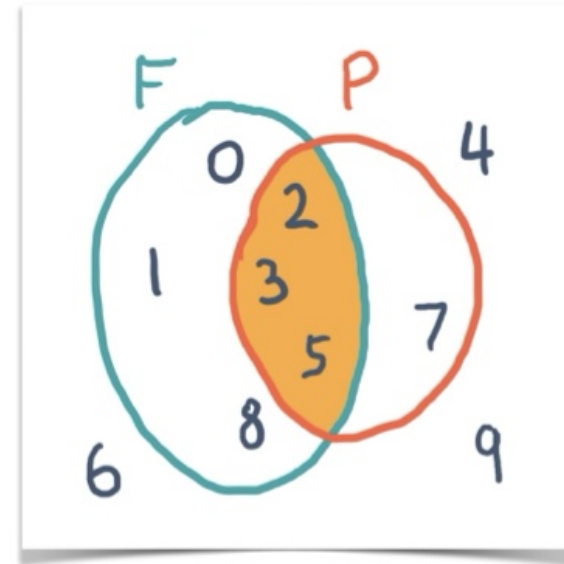
is the same as:

*x belongs to A **and**
x also belongs to B.*

Math notation:

$$x \in (A \cap B) \iff (x \in A) \wedge (x \in B)$$

In computing: **AND**



LOGIC DISJUNCTION: UNION

x belongs to the union of A and B.

is the same as:

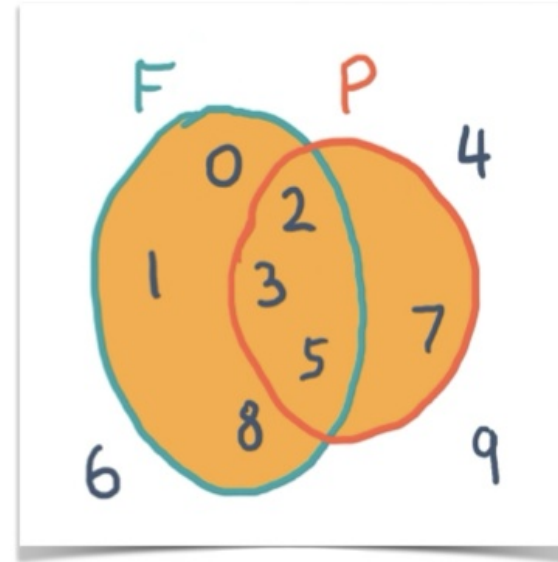
*x belongs to A **or***

x belongs to B.

Math notation:

$$x \in (A \cup B) \iff (x \in A) \vee (x \in B)$$

In computing: **OR**



DIFFERENCE

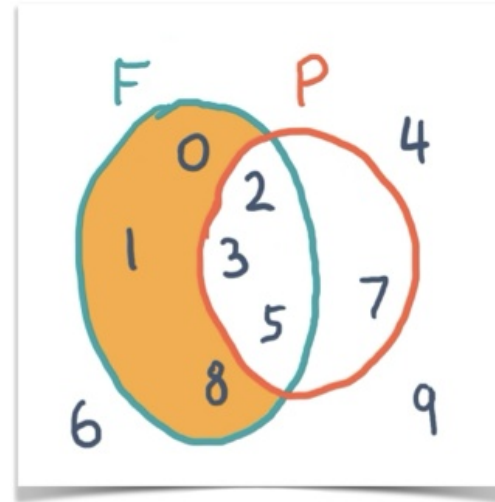
*x belongs to A but
does not belong to B .*

is the same as:

*elements of A minus
elements of B*

Math notation:

$$x \in (A \setminus B) \iff (x \in A) \wedge (x \notin B)$$



SYMMETRIC DIFFERENCE

*x belongs to A **or**
 x belongs to B but
does not belong to both*

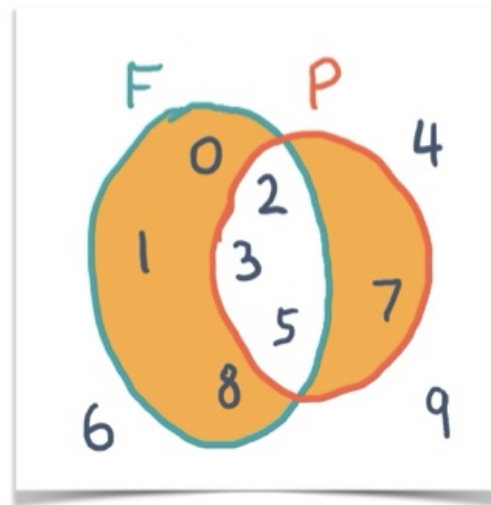
Is the same as:

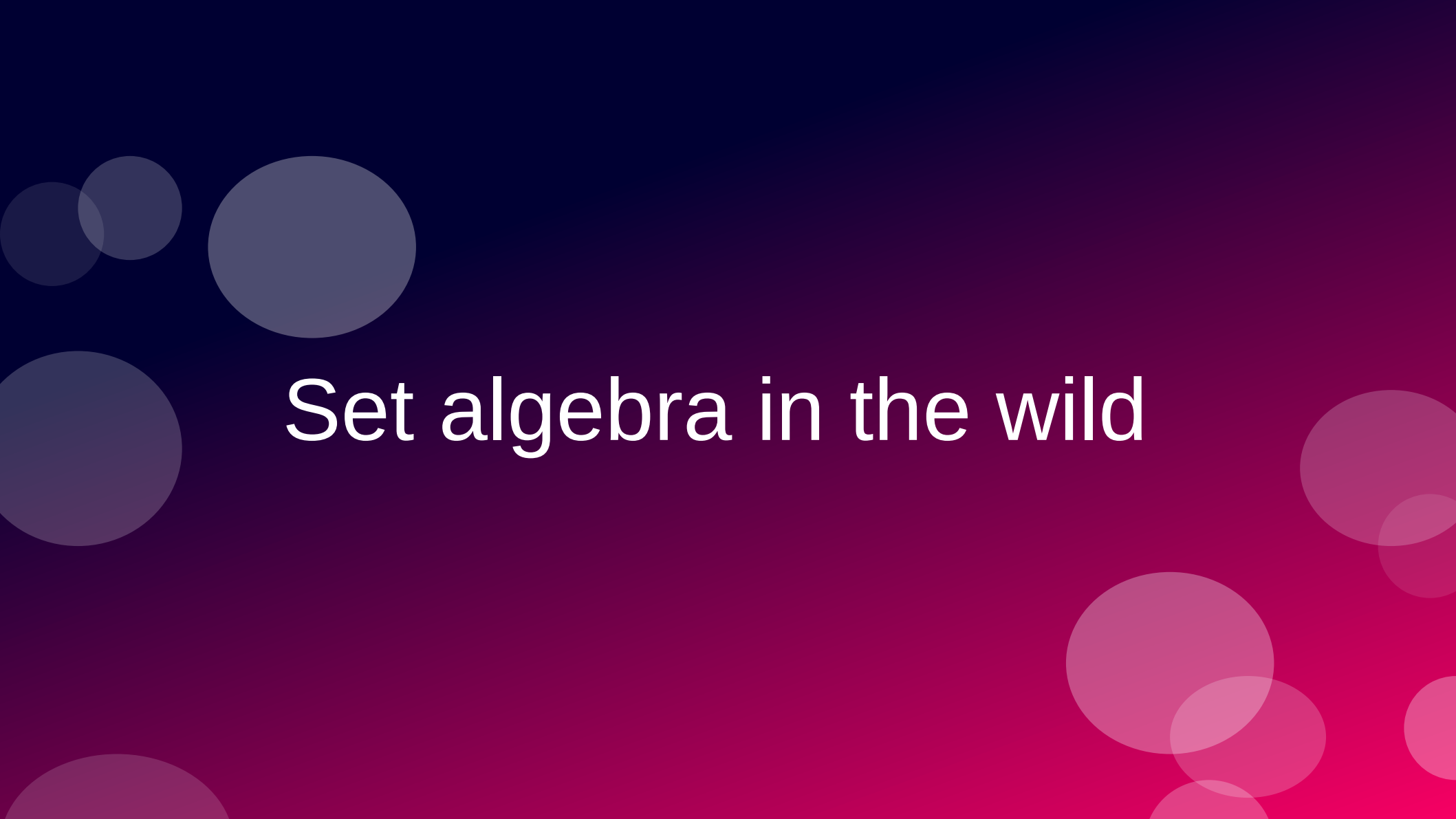
*x belongs to the union of A with B
less the intersection of A with B .*

Math notation:

$$x \in (A \Delta B) \iff (x \in A) \vee (x \in B)$$

In computing: **XOR**





Set algebra in the wild

Implementing sets

- Invariant: set elements are unique
 - Elements must be hashable and comparable
- $e \in S$ (membership test):
 - Expected to be $O(1)$ in a reasonable implementation

Set algebra

- Essential: $e \in S$
- Highly desirable:
 - $S \supseteq Z$ (superset; S contains all elements of Z)
 - $S \cap Z$ (intersection)
 - $S \cup Z$ (union)
 - $S \setminus Z$ (difference)

SETS IN A FEW LANGUAGES AND PLATFORMS

Some languages/platform APIs that implement sets in their standard libraries

Elixir	MapSet: 14 methods	😺
Ruby	Set: > 10 methods and operators	😺
Python	set, frozenset: > 10 methods and operators	😺
.Net (C# etc.)	ISet interface: > 10 methods; 2 implementations	😺
JavaScript (ES6)	Set: < 10 methods	
Java	Set interface: < 10 methods; 8 implementations	
Go	Do it yourself, or pick one of several packages...	

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SETS IN A FEW LANGUAGES AND PLATFORMS

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SETS IN A FEW LANGUAGES AND PLATFORMS

Some languages/platform APIs that implement sets in their standard libraries

Elixir	MapSet: 14 methods		} rich APIs
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JavaScript (ES6)	Set: < 10 methods		} lean APIs
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Evolution of set APIs

- Most methods in lean APIs handle single elements
 - e.g.: adding/removing an element, iteration
- “The von Neumann bottleneck”

Can Programming Be Liberated from the von Neumann Style? A Functional Style and Its Algebra of Programs

John Backus
IBM Research Laboratory, San Jose



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Author's address: 91 Saint Germain Ave., San Francisco, CA 94114.

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613

Conventional programming languages are growing ever more enormous, but not stronger. Inherent defects at the most basic level cause them to be both fat and weak: their primitive word-at-a-time style of programming inherited from their common ancestor—the von Neumann computer, their close coupling of semantics to state transitions, their division of programming into a world of expressions and a world of statements, their inability to effectively use powerful combining forms for building new programs from existing ones, and their lack of useful mathematical properties for reasoning about programs.

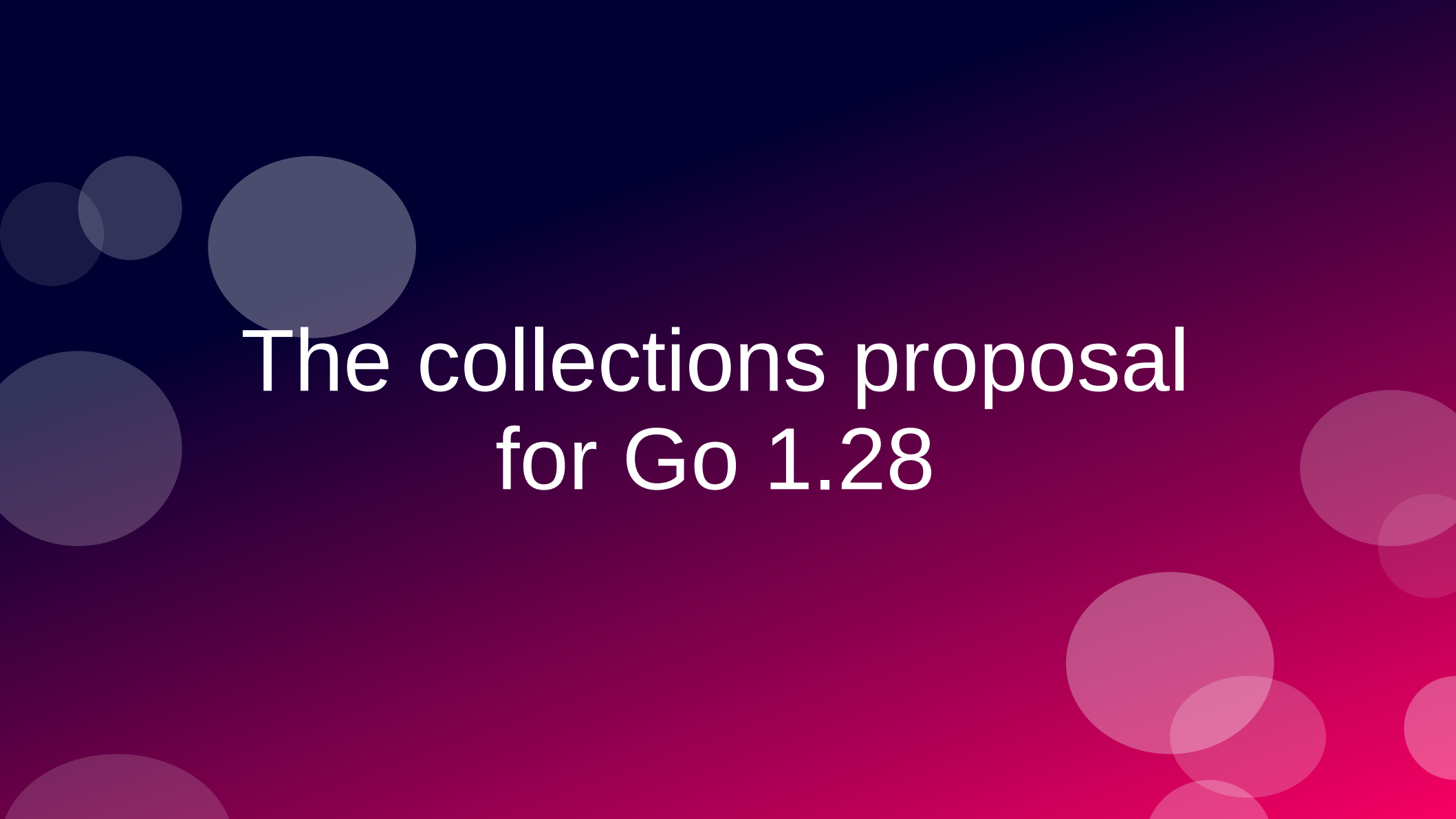
An alternative functional style of programming is founded on the use of combining forms for creating programs. Functional programs deal with structured data, are often nonrepetitive and nonrecursive, are hierarchically constructed, do not name their arguments, and do not require the complex machinery of procedure declarations to become generally applicable. Combining forms can use high level programs to build still higher level ones in a style not possible in conventional languages.

Communications
of
the ACM

August 1978
Volume 21
Number 8

Evolution of set APIs

- Languages are providing more set algebra operations
- ECMAScript 2025 added:
 - `Set.intersection`, `Set.union`,
 - `Set.difference`,
 - `Set.symmetricDifference`,
 - `Set.isSubsetOf`, `Set.isSupersetOf`,
 - `Set.isDisjointFrom`
- Go 1.28 will have a rich set API!



The collections proposal for Go 1.28

The Go Collections working group was formed in late 2025 with the purpose of bringing common collection data structures to the standard library, guided by the familiar Go principles of pragmatism and simplicity.

...

This work seeks to add several of the more important data types to the standard library, and to establish conventions for their APIs and those of future additions.

Alan Donovan
Go issue #80590

The 7 proposed additions

1) `hash/maphash.Hasher`

standard interface supporting custom hash functions and equivalence relations for arbitrary data types (released in Go 1.27)

2) `container/hash.Map[K, V]`

a hash-based Map that uses custom hash functions.

3) `container/hash.Set[T]`

a hash-based Set along the same lines.

4) `container/heap/v2.Heap`

a generic binary heap API to replace the standard library's existing heap (hard to use).

The 7 proposed additions

5) `container/set.Set[T]`

transparently represented as `map[T]struct{}`, supporting usual set operations such as Union and Intersection. We expect it to become the standard set in most new Go APIs.

6) `container/mapset`

helper functions (Union, Intersection, and so on) for manipulating legacy sets as sets in existing code whose API cannot be changed. `set.Set` wraps this API.

7) `container/ordered.Map[K, V]`

a map that preserves insertion order

proposal: container/...: generic collection types

- Umbrella issue:
<https://github.com/golang/go/issues/80590>
- 6 new concrete collection types
- 1 new interface Hasher (included in Go 1.27)
- 3 new abstract interfaces for collections, sets and maps

Source tree (WIP)

Open CIs
(unmerged)
identified
by #issue

go/src/	
└─ hash/	
└─ maphash/	
└─ maphash.go	= Hasher interface, in Go 1.27 (#70471)
└─ maps/	
└─ maps.go	★ added Identical (#78456), used by mapset
└─ container/	
└─ container_test.go	★+ abstract interfaces (unexported)
└─ set/	
└─ set.go	★+ the new canonical set (#69230)
└─ mapset/	
└─ mapset.go	★+ helpers for map-based sets (#77052)
└─ hash/	
└─ map.go	★+ hash.Map, uses maphash.Hasher (#69559)
└─ set.go	★+ hash.Set, same approach (#80584)
└─ heap/	
└─ heap.go	= existing, pre-generics
└─ v2/	
└─ heap.go	? generic, more ergonomic Heap (#77397)
└─ list/	
└─ list.go	= existing
└─ ordered/	
└─ ordered.go	? tree-based ordered.Map/Set (#60630)
└─ ring/	
└─ ring.go	= existing

Abstract interfaces

Unexported but
intended as models
for new collection types

```
go/src/  
├── hash/  
│   └── maphash/  
│       └── maphash.go  
├── maps/  
│   └── maps.go  
├── container/  
├── container/  
│   ├── container_test.go  
│   ├── set/  
│   │   └── set.go  
│   ├── mapset/  
│   │   └── mapset.go  
│   └── hash/  
│       ├── map.go  
│       └── set.go
```

Comment in container/container_test.go

```
• // The following interfaces define the abstract data types for
• // collections in Go. [...]
• // ...
• //
• // These interfaces are not yet published, but may be included in a
• // future Go release once we have experience of whether these methods
• // are necessary and sufficient.
• // ...
• // [interface definitions...]
• //
• // These types define the fundamental operations that need to be
• // implemented by all set and map types. Their naming, signature, and
• // semantic conventions should be followed wherever possible when
• // defining new collection types.
```

_AbstractCollection

```
1 // _AbstractCollection models a collection C of elements E,  
2 // such as *hash.Map, *hash.Set, *ordered.Map, or set.Set.  
3 type _AbstractCollection[E any, C _AbstractCollection[E, C]] interface {  
4     Clear()  
5     Clone() C  
6     Contains(E) bool  
7     ContainsAll(iter.Seq[E]) bool  
8     Len() int  
9     String() string  
10 }
```

Both _AbstractSet and _AbstractMap
embed _AbstractCollection

_AbstractSet

```
1 // _AbstractSet models a set S of elements E,  
2 // such as *hash.Set, or set.Set.  
3 type _AbstractSet[E any, S _AbstractSet[E, S]] interface {  
4     _AbstractCollection[E, S]  
5  
6     Insert(E) bool  
7     InsertAll(iter.Seq[E]) bool  
8     Equal(S) bool  
9     All() iter.Seq[E]  
10    Delete(E) bool  
11    DeleteAll(iter.Seq[E]) bool  
12    DeleteFunc(func(E) bool) bool  
13    Intersection(S) S  
14    IntersectionWith(S)  
15    Intersects(S) bool  
16    Union(S) S  
17    UnionWith(S)  
18    Difference(S) S  
19    DifferenceWith(S)  
20    SymmetricDifference(S) S  
21    SymmetricDifferenceWith(S)  
22 }
```

```
1 // _AbstractSet models a set S of elements E,  
2 // such as *hash.Set, or set.Set.  
3 type _AbstractSet[E any, S _AbstractSet[E, S]] interface {  
4     _AbstractCollection[E, S]  
5  
6     Insert(E) bool  
7     InsertAll(iter.Seq[E]) bool  
8     Equal(S) bool  
9     All() iter.Seq[E]  
10    Delete(E) bool  
11    DeleteAll(iter.Seq[E]) bool  
12    DeleteFunc(func(E) bool) bool  
13    Intersection(S) S  
14    IntersectionWith(S)  
15    Intersects(S) bool  
16    Union(S) S  
17    UnionWith(S)  
18    Difference(S) S  
19    DifferenceWith(S)  
20    SymmetricDifference(S) S  
21    SymmetricDifferenceWith(S)  
22 }
```



set algebra

_AbstractMap

```
1 // _AbstractMap models a mapping M from keys K to values V,  
2 // such as *hash.Map or *ordered.Map.  
3 type _AbstractMap[K, V any, M _AbstractMap[K, V, M]] interface {  
4     _AbstractCollection[K, M]  
5  
6     Set(K, V) (V, bool)  
7     SetAll(iter.Seq2[K, V]) bool  
8     Get(K) (V, bool)  
9     At(K) V  
10    All() iter.Seq2[K, V]  
11    Keys() iter.Seq[K]  
12    Values() iter.Seq[V]  
13    Delete(K) (V, bool)  
14    DeleteAll(iter.Seq[K]) bool  
15    DeleteFunc(func(K, V) bool) bool  
16 }
```

Static tests in container_test.go

```
1 // -- conformance --
2
3 // This is a static compilation test of various symmetries,
4 // expressed using F-bounded polymorphic interfaces to
5 // achieve covariant parameter/result specialization.
6
7 var _ _AbstractSet[int, set.Set[int]] = make(set.Set[int])
8
9 var _ _AbstractSet[int, *hash.Set[int]] = new(hash.Set[int])
```

The compiler allows assigning `set.Set` and `hash.Set` to a variable `_` of type `_AbstractSet`

Generic function in container_test.go

```
1 // ContainsAny reports whether set x contains any element of sequence y
2 func ContainsAny[E any, S _AbstractSet[E, S]](x S, y iter.Seq[E]) bool {
3     for elem := range y {
4         if x.Contains(elem) {
5             return true
6         }
7     }
8     return false
9 }
```

ContainsAny works with any _AbstractSet

Another generic function in container_test.go

```
1 // Take removes and returns an arbitrary element from a set.
2 // It returns zero if the set was empty.
3 func Take[S _AbstractSet[E, S], E any](set S) (e E, found bool) {
4     for e = range set.All() {
5         found = true
6         set.Delete(e) // may fail for NaN
7         break
8     }
9     return
10 }
```

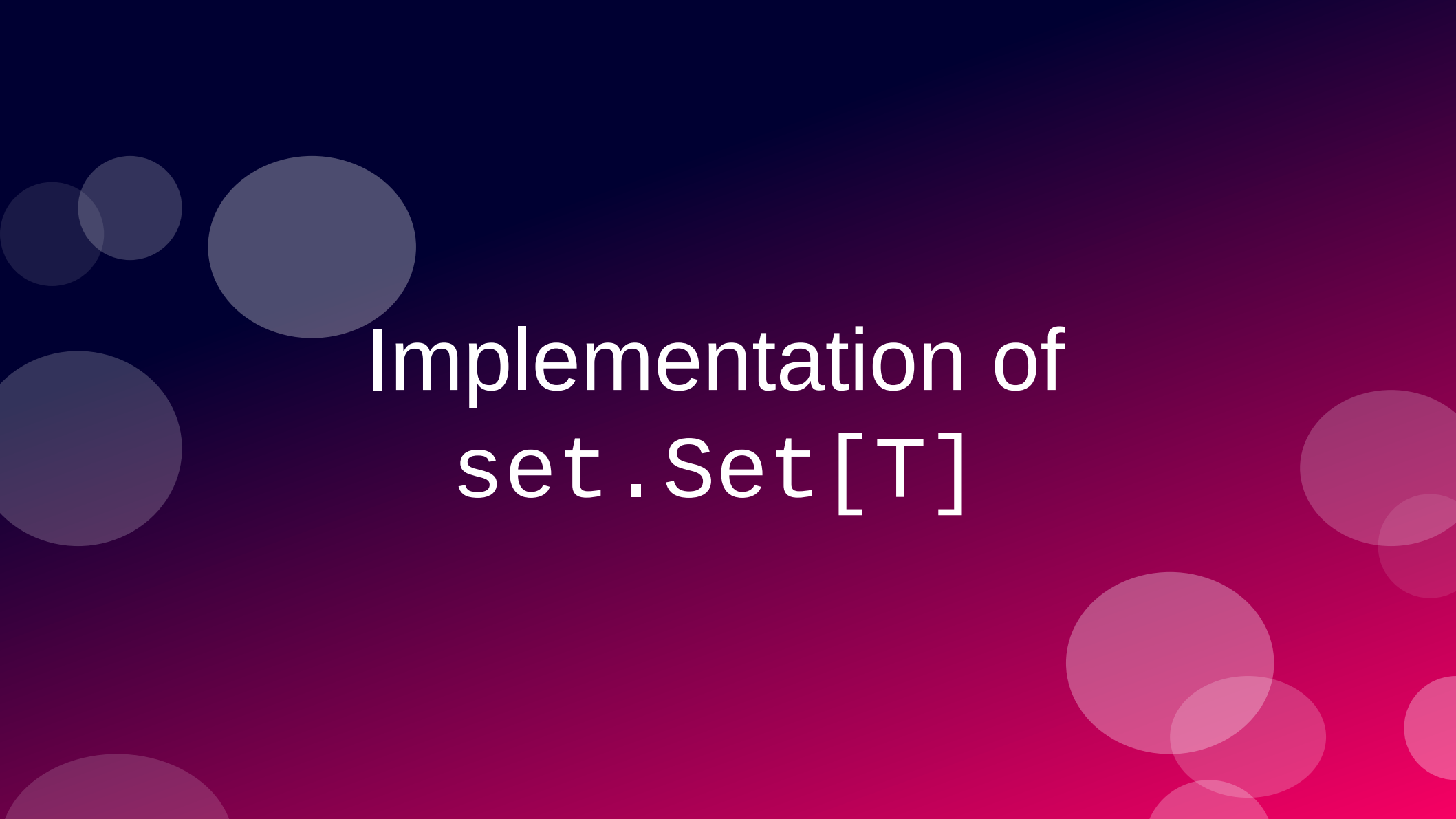
Take returns an element of type E
(from `_AbstractSet[E, S]`) and a bool

More set algebra, also in container_test.go

```
1 // Subset reports whether set x is a subset of set y.
2 func Subset[S _AbstractSet[E, S], E any](x, y S) bool {
3     // We cannot shortcut if x == y here because
4     // it may panic for some types (e.g. maps) or give
5     // the wrong answer for others (e.g. strings
6     // considered as unordered sets of bytes).
7     // Secondly, pointer identity also doesn't
8     // repeat NaN != NaN.
9
10    return x.Len() <= y.Len() && y.ContainsAll(x.All())
11 }
12
13 // Superset reports whether x is a superset of y.
14 func Superset[S _AbstractSet[E, S], E any](x, y S) bool {
15     return Subset(y, x)
16 }
```

$$x \subseteq y$$

$$x \supseteq y$$



Implementation of `set.Set[T]`

set.Set[T]

```
go/src/
├── hash/
│   └── maphash/
│       └── maphash.go
├── maps/
│   └── maps.go
├── container/
├── container/
│   ├── container_test.go
│   ├── set/
│   │   └── set.go
│   ├── mapset/
│   │   └── mapset.go
│   └── hash/
│       ├── map.go
│       └── set.go
```

set.Set delegates all work to mapset functions

```
1 package set
2
3 import (
4     "github.com/ramalho/sets-in-go/vendored/mapset" // proposed mapset
5     "iter"
6     "maps"
7 )
8
9 // A Set[E] is a set of elements of type E.
10 type Set[E comparable] map[E]struct{}
11
12 // Collect creates a new set containing the elements of the sequence.
13 func Collect[E comparable](seq iter.Seq[E]) Set[E] {
14     return Set[E](mapset.Collect(seq))
15 }
```

All but one of the functions and methods in set.go
are one-liners calling functions in mapset

Functions for using legacy maps as sets

```
go/src/  
├── hash/  
│   └── maphash/  
│       └── maphash.go  
├── maps/  
│   └── maps.go  
├── container/  
├── container/  
│   ├── container_test.go  
│   ├── set/  
│   │   └── set.go  
│   ├── mapset/  
│   │   └── mapset.go  
│   └── hash/  
│       ├── map.go  
│       └── set.go
```

The underlying mapset (uses no named types)

```
1 package mapset
2
3 import (
4     "iter"
5 )
6
7 // Collect returns a new set containing the elements of the sequence.
8 func Collect[K comparable](seq iter.Seq[K]) map[K]struct{} {
9     return collect[K, struct{}](seq)
10 }
11
12 /// ...
13
14 func collect[K comparable, V bool | struct{}](seq iter.Seq[K]) map[K]V {
15     x := make(map[K]V)
16     InsertAll(x, seq)
17     return x
18 }
```


mapset.InsertAll and mapset.insert

```
1 // InsertAll adds each element of the addenda sequence to the set.
2 // If the set values are boolean, the value 'true' is used.
3 // It reports whether len(x) changed.
4 func InsertAll[M ~map[K]V, K comparable, V bool | struct{}](x M, addenda iter.Seq[K]) bool {
5     pre := len(x)
6     for k := range addenda {
7         insert(x, k)
8     }
9     return len(x) != pre
10 }
11
12 func insert[M ~map[K]V, K comparable, V bool | struct{}](m M, k K) {
13     // Choose the distinguished "present" value (true or struct{}{}).
14     // This compiles to a load from .rodata.
15     var present V
16     if _, ok := any(present).(bool); ok {
17         present = any(true).(V)
18     }
19
20     // This is the canonical insertion operation.
21     // All maps created by this API use only the
22     // distinguished 'present' value for the result type.
23     m[k] = present
24 }
```

Hasher interface

```
go/src/
├── hash/
│   └── maphash/
│       └── maphash.go
├── maps/
│   └── maps.go
├── container/
├── container/
│   ├── container_test.go
│   ├── set/
│   │   └── set.go
│   ├── mapset/
│   │   └── mapset.go
│   └── hash/
│       ├── map.go
│       └── set.go
```

Set type with support for custom hashers

```
go/src/  
├── hash/  
│   └── maphash/  
│       └── maphash.go  
├── maps/  
│   └── maps.go  
├── container/  
├── container/  
│   ├── container_test.go  
│   ├── set/  
│   │   └── set.go  
│   ├── mapset/  
│   │   └── mapset.go  
│   └── hash/  
│       ├── map.go  
│       └── set.go
```



Wrapping up

Key takeaways

- Set algebra provides simpler, faster solutions to common data processing tasks.
- Sets and other collections in the upcoming Go 1.28 provide a foundation for the design of generic collections.
- The Go source code shown here is unmerged and may change!

Mandatory AI tips

- For developers using agents:
The precise vocabulary of set algebra is useful to prompt coding agents, regardless of the programming language.
- For citizen programmers:
Learning set algebra and the relational model may be the best foundation for agentic coding.
- For everyone:
Demand your time back. Time is not money, it is your time to live!

Remember Aaron Schwartz

- Co-creator of RSS and Markdown
- Co-founder of Reddit
- Aaron was arrested for downloading scientific articles from JSTOR in bulk
- His life was destroyed by the Obama DOJ
- He never shared the articles with anyone.
- What would he would do with them?
Perhaps train a model?



The Knowledge Smoothie

SAM ALTMAN: Let's make a smoothie!
Who can contribute with fruits?

EVERYONE: OK, here's the fruit I have!

SAM ALTMAN: The smoothie is ready.
It's \$20 a glass!

EVERYONE: But you took our fruits to
make it!

SAM ALTMAN: No I didn't! Where's your
fruit? Show me!



The Solution

- Any LLM vendor who refuses to disclose every external source used in training must open source the model and its weights.
- Too radical?
- It's only fair and common sense.

